CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 4: Quadratic Equations

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Check whether $(x+2)(x-3) = (x-1)(x+4)$ is a quadratic equation.
2. Solve by factorisation: $x^2 - 3x - 10 = 0$.
3. Solve by factorisation: $2x^2 - 9x + 10 = 0$.
4. Solve by factorisation: $6x^2 + 5x - 6 = 0$.
5. Solve using the quadratic formula: $x^2 - 4x + 1 = 0$.
6. Find the discriminant of $3x^2 - 5x + 2 = 0$ and state the nature of its roots.
7. Find the value of k for which $2x^2 + kx + 8 = 0$ has equal roots.
8. Find two consecutive positive integers whose squares add up to 313.
9. The altitude of a right-angled triangle is 7 cm less than its base. If the hypotenuse is 17 cm, find the other two sides.
10. A motorboat whose speed in still water is 15 km/h goes 30 km downstream and comes back in a total of 4 hours 30 minutes. Find the speed of the stream.

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Answer Key

1. It simplifies to $4x + 2 = 0$, so it is not quadratic.
2. $x = 5$ or $x = -2$
3. $x = \frac{5}{2}$ or $x = 2$
4. $x = \frac{2}{3}$ or $x = -\frac{3}{2}$
5. $x = 2 \pm \sqrt{3}$
6. $D = 25 - 24 = 1 > 0$: two distinct real roots
7. $k^2 = 64$, so $k = \pm 8$
8. 12 and 13
9. Base 15 cm, altitude 8 cm
10. 5 km/h

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